Case Study Report: Troubleshooting Random Workstation Restarts Under Heavy Load

Scenario

At **Nexus Innovations**, three rendering workstations successfully complete the morning **Power-On Self-Test (POST)** but begin to restart randomly during heavy rendering workloads in the afternoon. Investigation reveals that all three systems are placed inside poorly ventilated desk cabinets, restricting airflow and causing heat buildup.

Objective

To identify the probable hardware causes of random system restarts, explain the reason behind the failures, diagnose the issue using stress-testing tools, recommend corrective actions, and determine whether upgrading the Power Supply Unit (PSU) is necessary.

1. Hardware Components Most Likely Responsible for Random Reboots

The following hardware components are the most probable causes of unexpected restarts under heavy workloads.

A. Central Processing Unit (CPU)

The CPU performs complex rendering calculations that generate significant heat. If the processor temperature exceeds its safe operating limit, the motherboard activates thermal protection to prevent permanent damage by shutting down or restarting the computer.

Possible Symptoms:

- High CPU temperature
- Sudden restart during rendering
- Reduced performance due to thermal throttling

B. Power Supply Unit (PSU)

During rendering, the CPU, GPU, memory, and storage devices consume more electrical power. If the PSU cannot provide sufficient and stable power, voltage fluctuations may occur, causing the computer to restart unexpectedly.

Possible Symptoms:

- Random restarts only during heavy workloads
- System instability
- Unexpected shutdowns without warning

C. Cooling System (CPU Cooler and Case Fans)

The cooling system removes heat generated by internal components. Dust accumulation, failed fans, dried thermal paste, or restricted airflow inside enclosed cabinets reduce cooling efficiency and increase internal temperatures.

Possible Symptoms:

- High internal temperatures
- Loud fan noise
- Frequent thermal shutdowns

2. Why the Systems Pass POST in the Morning but Fail in the Afternoon

The **Power-On Self-Test (POST)** is performed immediately after the computer is switched on. During the morning:

- The room temperature is relatively low.
- Internal hardware components are cool.
- The CPU has not yet been placed under heavy workload.
- Cooling systems can easily maintain safe temperatures.

Therefore, the computer successfully passes POST.

During the afternoon:

- Rendering software places the CPU and GPU under continuous heavy load.
- Heat accumulates inside the poorly ventilated desk cabinets.
- Restricted airflow prevents hot air from escaping.
- Internal temperatures continue increasing.
- Once the CPU or other components exceed their thermal limits, protective mechanisms automatically restart the computer to prevent hardware damage.

Thus, the systems operate normally when cold but become unstable after prolonged operation.

3. Using a Stress-Test Tool to Confirm a Thermal or Power Issue

A stress-test simulates maximum hardware workload to determine whether the system remains stable.

Procedure

Step 1

Install a stress-testing application such as:

- Prime95
- AIDA64
- OCCT
- Cinebench (CPU Benchmark)

Step 2

Install hardware monitoring software such as:

- HWMonitor
- HWiNFO
- Open Hardware Monitor

Step 3

Record the following values before starting the test:

- CPU Temperature
- GPU Temperature
- Fan Speed
- CPU Frequency
- Voltage Readings

Step 4

Run the stress test for approximately **20–30 minutes** while monitoring hardware temperatures.

Step 5

Observe the results.

If:

- CPU temperature rises above approximately **90°C**
- GPU temperature becomes excessively high

- Voltage becomes unstable
- The computer suddenly restarts

then the problem is confirmed as either a thermal issue or an insufficient power supply.

4. Recommended Solutions to Prevent Frequent System Restarts

Solution 1: Improve Airflow and Ventilation

Remove the workstations from enclosed desk cabinets or provide adequate ventilation.

This allows:

- Better airflow
- Lower internal temperatures
- Improved cooling efficiency

Solution 2: Service the Cooling System

Perform preventive maintenance by:

- Cleaning dust from heatsinks and fans
- Replacing dried thermal paste
- Ensuring all cooling fans operate correctly
- Installing additional case fans if required

This improves heat dissipation and prevents thermal shutdown.

Solution 3: Test and Replace Faulty Hardware

Perform hardware diagnostics on:

- PSU
- CPU Cooler
- Motherboard
- RAM

Replace any component that fails diagnostic testing.

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5. Technical Justification for a PSU Upgrade

A PSU upgrade should only be recommended after verifying the existing power supply.

When a PSU Upgrade is Necessary

Upgrade the PSU if:

- Power consumption exceeds PSU capacity.
- Voltage output is unstable.
- The PSU is old or showing signs of failure.
- A powerful GPU or CPU has recently been installed.

A higher-quality PSU provides stable voltage and sufficient current under heavy workloads, reducing unexpected restarts.

When a PSU Upgrade is Not Necessary

A PSU upgrade is unnecessary if:

- Voltage remains stable during stress testing.
- The PSU wattage is sufficient.
- The restarts are caused solely by overheating due to poor airflow.

In this situation, improving cooling and ventilation is a more effective solution than replacing the PSU.

Observation

- The workstations passed POST successfully when temperatures were low.
- Continuous rendering increased CPU and internal cabinet temperatures.
- Poor airflow caused excessive heat accumulation.
- Stress testing reproduced the restart issue under heavy load.
- Improving cooling significantly reduced system instability.

The issue was successfully diagnosed through hardware inspection and stress testing. After improving airflow, servicing the cooling system, and verifying power supply performance, the workstations operated normally under heavy rendering workloads without random restarts, successfully meeting the required performance and stability criteria.